

Week 2

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2026-01-20

```
library(astsa)
```

```
## Warning: package 'astsa' was built under R version 4.5.2
```

```
library(xts)
```

```
## Warning: package 'xts' was built under R version 4.5.2
```

```
## Loading required package: zoo
```

```
## Warning: package 'zoo' was built under R version 4.5.2
```

```
##
```

```
## Attaching package: 'zoo'
```

```
## The following objects are masked from 'package:base':
```

```
##
```

```
##      as.Date, as.Date.numeric
```

```
#View(djia)
```

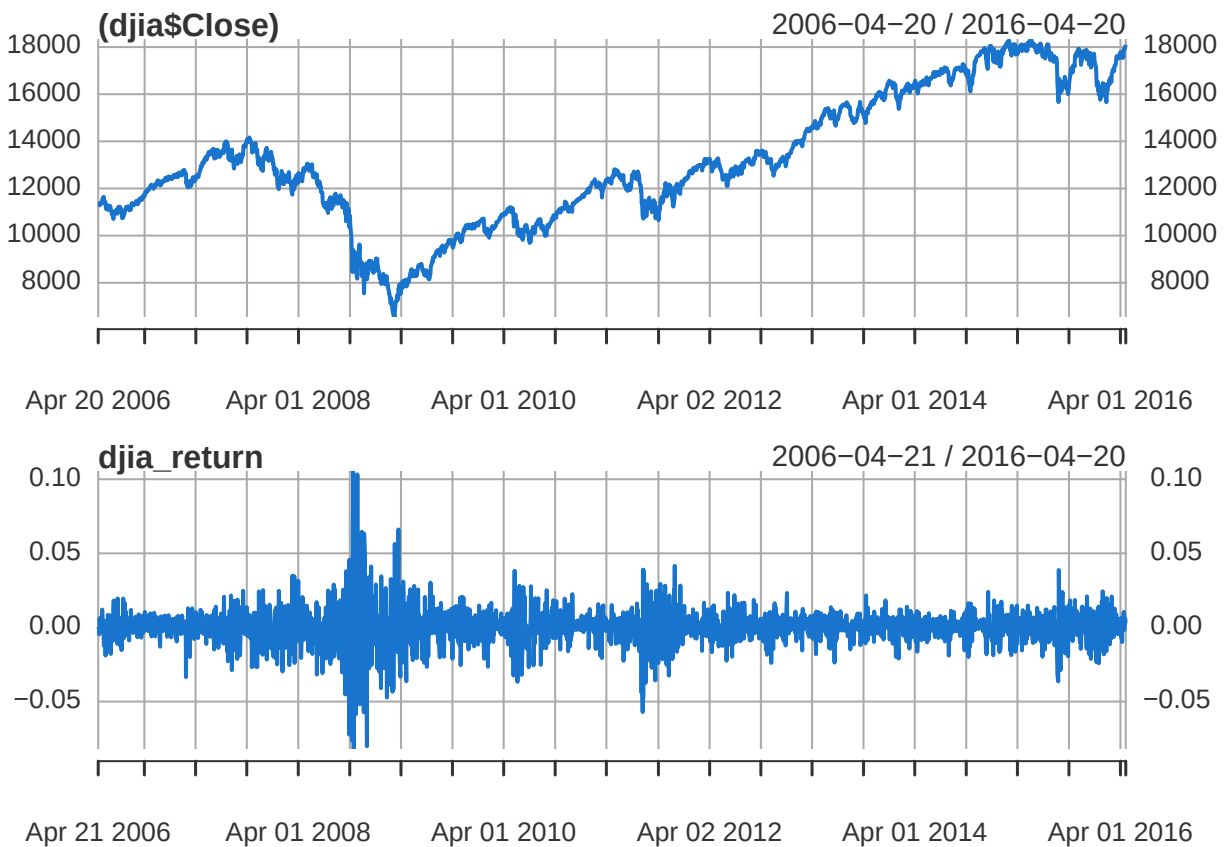
$$\log(1 + r_t) \approx r_t$$

```
djia_return = diff(log(djia$Close))[-1]  
djia_return
```

```
##              Close  
## 2006-04-21  0.0004019814  
## 2006-04-24 -0.0009813081  
## 2006-04-25 -0.0046924330  
## 2006-04-26  0.0062939559  
## 2006-04-27  0.0024646652  
## 2006-04-28 -0.0013512402  
## 2006-05-01 -0.0021003229  
## 2006-05-02  0.0064289317  
## 2006-05-03 -0.0014173744  
## 2006-05-04  0.0033784210
```

```
##      ...
## 2016-04-07 -0.0098752737
## 2016-04-08  0.0019932281
## 2016-04-11 -0.0011698726
## 2016-04-12  0.0093453502
## 2016-04-13  0.0104986520
## 2016-04-14  0.0010130063
## 2016-04-15 -0.0016172870
## 2016-04-18  0.0059439952
## 2016-04-19  0.0027422382
## 2016-04-20  0.0023607246
```

```
par(mfrow= c(2,1))
plot((djia$Close), col = 4)
plot(djia_return, col = 4)
```



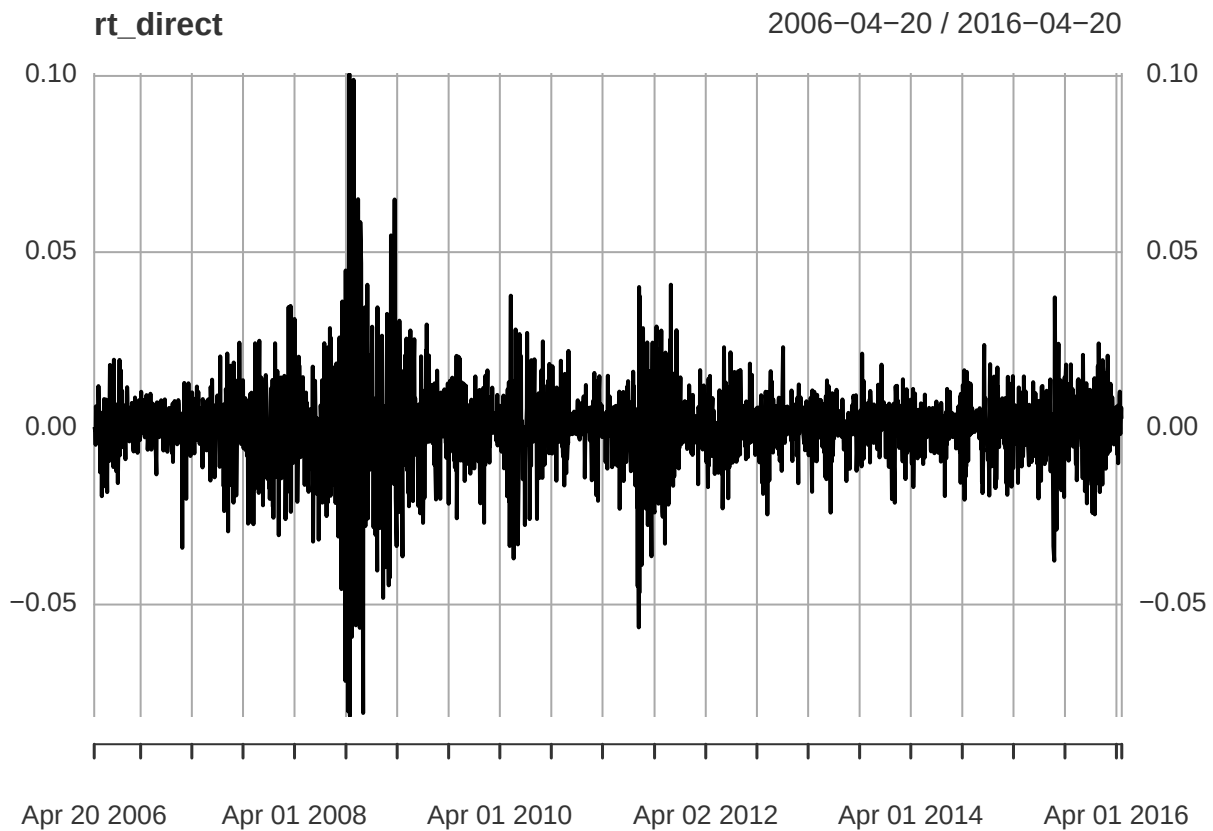
$$r_t = \frac{x_t - x_{t-1}}{x_{t-1}}$$

```
rt_direct = diff(djia$Close)/lag(djia$Close, -1)
cbind(rt_direct, djia_return)
```

```
##      Close      Close.1
## 2006-04-20      NA      NA
## 2006-04-21 0.0004022952 0.0004019814
```

```
## 2006-04-24 -0.0009864076 -0.0009813081
## 2006-04-25 -0.0046739493 -0.0046924330
## 2006-04-26 0.0062587457 0.0062939559
## 2006-04-27 0.0024649589 0.0024646652
## 2006-04-28 -0.0013549965 -0.0013512402
## 2006-05-01 -0.0020890565 -0.0021003229
## 2006-05-02 0.0064173998 0.0064289317
## 2006-05-03 -0.0014135955 -0.0014173744
## ...
## 2016-04-07 -0.0099044336 -0.0098752737
## 2016-04-08 0.0019935738 0.0019932281
## 2016-04-11 -0.0011596689 -0.0011698726
## 2016-04-12 0.0092046724 0.0093453502
## 2016-04-13 0.0104331593 0.0104986520
## 2016-04-14 0.0010141322 0.0010130063
## 2016-04-15 -0.0016090031 -0.0016172870
## 2016-04-18 0.0059101354 0.0059439952
## 2016-04-19 0.0027320246 0.0027422382
## 2016-04-20 NA 0.0023607246
```

```
plot(rt_direct)
```



```
#The Lag is similar to  $X_{t-1}$ 
```

```
difference <- rt_direct - djia_return  
difference <- na.omit(difference)  
mean(difference)
```

```
## [1] -5.869179e-05
```

We can perform χ^2 - test and t - test to compare the standard deviation and mean.